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Exploratory Data Analysis of NFL Player Performance and Impact of Age on Speed and Acceleration

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***Abstract*—This study examines the role of speed and agility in shaping player performance and game strategies in American football. By investigating the relationship between speed, acceleration, and age, the study uncovers trends that shed light on what makes players effective on the field. A comprehensive data analysis process, including preprocessing, feature extraction, and normalization, ensures a robust examination of these relationships. The findings highlight the complex dynamics between physical attributes and age, revealing insights into player development and performance dynamics. While offering valuable insights, the study also emphasizes the need for further investigation to account for outliers and refine the understanding of these critical performance factors. This study lays a solid foundation for improving how we evaluate and predict player potential, making a valuable contribution to performance analysis in football.**

Keywords— pandas, numpy, seaborn, matplotlib, exploratory data analysis, NFL, physical predisposition

I. Introduction

Football is one of America's most popular, if not the most popular sports. As with every other sport, it is constantly evolving in many different ways, from the technical presentation of games for fans to the actual players who physically become more dominant, their lifespan as a player increases but they also physically change. The result of these changes is that this sport nowadays requires players who are physically prepared to deliver the best possible performances day in and day out. Two of the most important characteristics of the players in the 21st century from the physical perspective are [1]:

- Physical/muscle strength
- Agile capabilities (speed, agility, acceleration, etc.)

In this paper, the focus is placed on the second characteristic of players, speed, and agility and how this specific characteristic influences the performance of teams but also how age influences players' capabilities of speed and acceleration.

Throughout the history of this sport, a lot of statistical information has been collected on the multiple different levels of this sport (university level, high school level, professional level, etc.) which helps assume the ability of players but also predict outcomes of the matches which highly influenced gambling industry [1]. Other than the gambling industry, statistical analysis of every player, every game, and every play, also highly influences tactics for the future game and since this is a type of sport where tactics and right calls decide the winner and loser, it is highly important to have good analysis. Additionally, a thorough analysis can help sport managers as well as sport scouts, to find suitable players, make valuable decisions regarding transfers and others. In recent years, the integration of data analytics in football has significantly improved the recruitment process. Clubs now use detailed performance metrics to identify players whose skills align with their tactical requirements. For instance, Brentford FC has employed data-driven models to discover talent in various leagues, leading to the acquisition of players like Ollie Watkins and Neal Maupay, who were later sold for substantial profits. Similarly, Liverpool FC's data-centric approach has been essential in assembling a squad that secured the Champions League title in 2019 and the Premier League in 2020. Their recruitment strategy, informed by advanced analytics, facilitated the signings of key players such as Mohamed Salah, Virgil van Dijk, and Alisson Becker [14].

The goal of this paper is to examine how speed and agility influence team performance and to assess the impact of age on players' speed and acceleration capabilities.

II. Literature review

The integration of big data analytics and advanced computational technologies has significantly improved how athlete performance is evaluated, particularly in high-impact sports like the National Football League (NFL). Understanding how age influences key physical attributes such as speed and acceleration is critical for player development, team strategy, and injury prevention. Recent research has leveraged a combination of machine learning, cloud computing, and data modeling techniques to explore these relationships in depth.

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Watanabe et al. [15] highlight the transformative role of big data in sports management, emphasizing its impact on decision-making through predictive analytics and machine learning algorithms. By utilizing large-scale datasets and AI-driven models, teams can assess how player performance changes over time, particularly in terms of speed and acceleration. These insights help refine training regimens and optimize player utilization strategies.

Ma [16] explores how big data analytics is used in sports education, with an emphasis on data-driven models that evaluate different facets of athlete performance. The study analyzes physical characteristics and skill development using statistical modeling and machine learning techniques. These models offer insightful information about how aging affects athletic performance, enabling customized training plans that counteract age-related decreases in acceleration and speed.

Zhang et al. [17] explore the feasibility of leveraging big data and IoT (Internet of Things) technologies in physical training. By integrating wearable sensors and motion tracking systems, this research enables real-time monitoring of player metrics. The study also utilizes cloud-based data storage and deep learning algorithms to analyze long-term performance trends, offering a more dynamic approach to understanding age-related changes in player speed and acceleration.

A sports prediction model that combines big data analytics, cloud computing, and artificial intelligence is presented by Eid et al. [18]. This model predicts performance differences between age groups by processing plant datasets using machine learning algorithms. The study helps coaches make data-driven judgments by integrating real-time data processing with AI-driven pattern recognition to deliver insightful predictions on how aging impacts acceleration and speed.

When taken as a whole, these studies demonstrate how new technologies like artificial intelligence (AI), machine learning, cloud computing, and the Internet of Things (IoT) are transforming the way that athlete performance is analyzed. Researchers and sports analysts can obtain a thorough grasp of how ageing affects NFL players' speed and acceleration by utilizing these resources. Enhancing training methods, prolonging career longevity, and raising overall player performance in professional football are all made possible by this understanding.

III. Methods and Materials

A. Dataset Overview

The dataset used in this research is a dataset provided directly by the National Football League to have their annual Big Data Bowl competition. The name of the dataset is NFL Big Data Bowl 2023 [2] and the goal of this competition is to use Next Gen Stats [3] player tracking data to generate some kind of statistics or to use data patterns to find out potentially useful data points or attributes which can be used in improving this sport.

Dataset consists of 12 different .csv files where 8 of them are first 8 weeks of games where every single play is captured with corresponding data which you can see in the Table I.

TABLE I. Attributes in the week 1-8 csv files

Column	Description
gameId	Unique identifier of all games
playId	Unique identifier of all plays during the game
nflId	Unique identifier of players
frameId	Frame identifier of every play (starts from 1)
time	Play timestamp
jerseyNumber	Jersey number of players
club	Player's club abbreviation
playDirection	Is play directed to left or right (which team is attacking)
x	Players position on X axis
y	Players position on Y axis
s	Players speed in yards/second
a	Player's acceleration in yards/second^2
dis	Distance traveled during the play

The topic for this specific edition of this competition is focussed on the offensive line that is protecting the pass of the team and also the opposing team that is working in the defensive line that is trying to prevent the passer from delivering the pass by either blocking the pass, stealing the ball or tackling the passer.

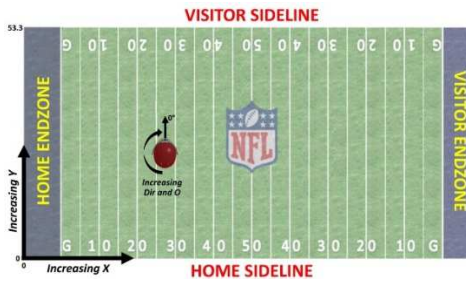

Fig. 1. Graphical representation of field

Based on Fig. 1. it is possible to see in what way parameters x and y are used when play is being described. Everything regarding plays works as a 2D coordinate system where the home team is on the left side and the visitor team is on the right side and based on the direction of the attack it is possible to conclude which team is attacking. The starting end of the coordinate system is always the endzone of the home team and based on the x and y coordinates together with the direction and orientation of the player, it is possible to create a full graph of the whole play frame by frame.

The second CSV file in the dataset is the games.csv file which only explains relevant information about the game and teams that are part of the game which you can see in Table II.

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TABLE II. ATTRIBUTES IN THE GAMES.CSV FILE

Column	Description
gameId	Unique identifier of all games
season	Year representation of the season that game took place in
week	In what week of the season game took place
gameDate	Exact game date
gameTimeEastern	Time at which game took place (Eastern time is used as time zone)
homeTeamAbbr	Abbreviation of home team
visitorteamAbbr	Abbreviation of visitor team

The third CSV file in the dataset is the pffScoutingData.csv file which consists of all the data that scouts provided for all players in their respective plays. This data explains certain abilities of the players. These attributes only explain how good or bad certain players performed in every play, not their general ability.

The fourth CSV file in the dataset is the players.csv file which consists of all players in the NFL with their ID, name, and physical measurements of their height and weight.

Fifth, and also last, the CSV file in the dataset is the plays.csv file which consists of more in-depth data about the movement of the players during plays. This file is not going to be covered that much in-depth since this data is not relevant to this research.

B. Data Analysis Techniques

Technologies that are used for this research are well-known libraries that are used in most of the machine learning-related research when we are talking about Python programming language: seaborn, pandas, matplotlib, and numpy.

Pandas played an important role in handling the datasets used for the research. Tasks such as merging multiple datasets, extracting player ages, and cleaning data were simplified through pandas. It allowed for efficient preprocessing by removing irrelevant entries, such as those with missing player identifiers, ensuring the dataset was well-prepared for analysis [4]. The NumPy was used for mathematical computations, particularly in calculating mean values for speed and acceleration. These mean values were crucial for normalizing the dataset, a process that assessed the uniformity in the data and improved its suitability for analysis. The library's support for multi-dimensional arrays and matrices provided the mathematical foundation required for this study [5]. For data visualization, Seaborn and matplotlib were used to present findings in a visually appealing and understandable manner. Seaborn was used to create detailed plots that revealed trends and patterns, such as the relationships between speed, acceleration, and age across different player groups. Its advanced visualization capabilities made it possible to display data distributions effectively, such as violin plots and heatmaps [6]. Matplotlib complemented seaborn by enabling the customization and creation of comparative and static plots, further aiding in the clear presentation of key metrics derived from the analysis [7].

C. Preprocessing Stage

Generally speaking, the dataset that is provided by the NFL for purposes of this competition is pretty prepared in most of the areas and it is providing great tools for data scientists to explore NFL data deeply. For these research purposes, there is some minor preprocessing needed to get insight into some of the data. The most notable preprocessing that needs to be done is extracting the age of the players and removing data points that do not reflect players' actions (data points where NFL is NaN).

The age of a player is straightforward when it comes to extractions since the dataset provides us with the full date of birth of every single player, so we can simply subtract the current date from the date of the player's birth, and we get the player's current age. The second important thing is to, first of all, merge all weeks of games into one big data frame because it makes life easier to work with one dataset instead of 8 different ones, and lastly remove all data points in the new big data frame where nflid (unique identifier of every player, refer to Table I. For a detailed explanation of all attributes of this specific CSV file.

Lastly, to get the most accurate data, some form of normalization of the dataset is needed.

The dataset is normalized in this case by calculating mean values of speed and acceleration which are going to be used in analyzing this dataset.

IV. RESULTS

In this section, the results will be presented through a series of visualizations, effectively illustrating player performance based on the analyzed attributes.

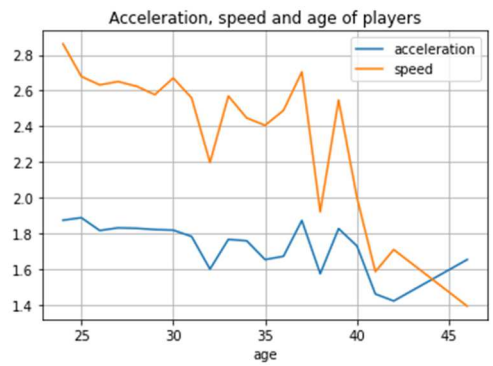

Fig. 2. Graph of players' acceleration and speed based on their age groups

In Fig. 2. it is easy to notice trends in how speed and acceleration change over time based on players' age groups. The premise before this research was that younger players are more agile (have greater speed and acceleration) and that attribute is dropping down slowly as they get older. From left to right on the graph, this is mostly accurate until we reach the age of 32 where there is a certain drop and immediately in the group of 33-year-old players, our trend continues. As we approach players in their 40s, more unexpected moments occur where at age 41 we can notice a certain drop in both stats and then at the age of 42, rise in speed. The last anomaly

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that is noticed here, compared to our premise, is that at the age of 46, there is quite a noticeable rise in players' acceleration that surpasses even younger players of age 32 which is not that expected if it is taken into consideration that most of the people expect that as players age, especially in the physically challenging sport like American Football, their speed and acceleration (and most of the others physical attributes) would be noticeably lower compared to players that are over 10 years younger than them [8].

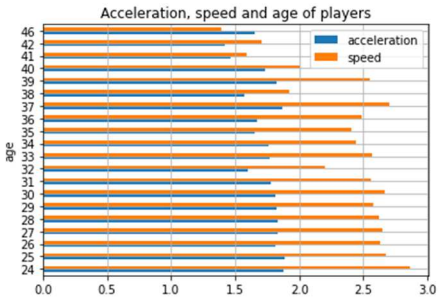

Fig. 4. Comparison of speed and acceleration by age (horizontal)

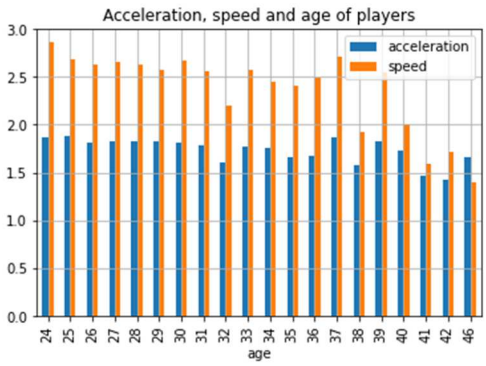

Fig. 3. Comparison of speed and acceleration by age (vertical)

In Fig. 3. and Fig. 4., it is possible to see additional views of how acceleration (blue bar) and speed (orange bar) differ based on the age groups of the players. Again, it is easy to notice a few discrepancies in some of the age groups like age 32, and most notably, age 46 which is showing much better acceleration than some of the players in younger age groups. Reasons for these types of anomalies differ and it is not easy to give a specific explanation why some older people simply have some better physical attributes, but it is believed that some groups of older people simply live healthier lifestyles and maintain their physique much better which gives them the ability to outperform younger colleagues [9]. This is even more notable when we are talking about sports people since it is not easy to play at the highest level in such old age like in our case (46), only players who sacrifice their life to reach peak performance can withstand this pressure. The second reason why acceleration might be a better attribute for some older players is that this attribute is not necessarily dependent only on the physical predispositions, but rather good technique and energy management which experienced players certainly have compared to players at the beginning of their careers.

Based on all the data gathered from Fig. 1., Fig. 2. and Fig. 3., it is clear that there is no real possibility to conclude how speed and acceleration are related to age since a lot of the outliers in some of the age groups are making the pattern unstable. To work around this issue, a different approach has been selected to deal with this problem.

A new grouping is going to be formed, where there won't be age groups with 1 year difference. Instead, there are going to be different age groups, with an increasing age of 4 between each of the groups. That means that the first group is for players with ages ranging from 24 to 27 (including 27). Based on this, there are going to be 5 different groups where the last group is going to be for all the players that are 40 years old and above since this is an age group where not a lot of players are active and they should be relatively similar when it comes to physical predispositions.

In Fig. 5., it is possible to see how many players are in each of the age groups. Most of the players are in the range from 28 to 31 (including 31), which is to be expected since

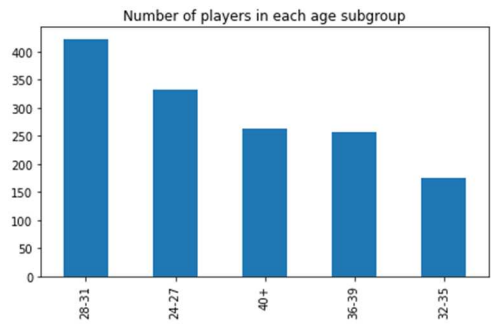

Fig. 5. Number of players in each age subgroup

this is the age group where players are in their prime (in most sports) so it is easy to guess that this is a group that has significantly more players than any other group. Interestingly, the group with the least number of players is from 32 to 35 (including 35). Looking at the dataset without having any deeper analysis, the premise would be that the age group of 40+ years old players will have the least amount of players since not a lot of players manage to play up until 4th decade let alone deep into that decade of their life since people in this age group do not have same physical ability as younger players so they require more special strategies when playing to stay relevant [10]. The reason why there are not many players in the age group of 32-35 is that this is the age group where most players that do not make the first roster are simply already retired. In fact, the average age of retirement for NFL players is around 27.5 but, as in all other sports of this kind, boundaries are being moved and players' lifespans are becoming more and more flexible so players that do have something to offer to this sport still manage to make it past that point [11]. This statistic also depends on the player's playing position since some positions are more physically demanding than others [12]. Lastly, pension plans for certain players can become quite tricky, so for players who do not earn top-tier salaries, pushing for a longer career is of big importance, even if that means playing for some minor team [13].

V. Conclusion

In this research paper, two different approaches to the analysis of how age impacts an NFL player's ability speed and acceleration are discussed. The initial premise is that age does make an impact on players in a way that as players age, their

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physical ability, which includes speed and acceleration, becomes worse over time.

In the first approach, where players were grouped in groups by their age, this premise was not confirmed. There was a contradiction to this premise because there were quite a lot of outliers in some of the older age groups, which ended up showing that some of the players in the older groups have better acceleration and speed abilities than their younger colleagues.

In this paper, a second approach was taken to check if the initial premise is correct. This second approach relies on dividing all the players into smaller subgroups that are going to take into account the age ranges of players of similar age where it is expected that all these players, due to their similar age, have similar physical capabilities. This approach is taken to deal with all the outliers and spread out physical abilities in a more even way.

Comparing these results to previous research, Watanabe et al. (2021) highlight the use of predictive analytics in sports management, which supports the idea that factors other than age affect player performance. Ma (2024) supports the necessity for improved approaches in subsequent investigations by highlighting the efficacy of machine learning models in performance assessment. According to Zhang et al. (2023), wearable technology and the Internet of Things can be used to measure performance parameters in real time, which could improve the precision of age-related performance evaluations. Eid et al. (2024) support the notion that an athlete's skill is not solely determined by age by showcasing how AI-driven sports prediction algorithms may predict performance reductions.

However, by directly comparing individual player speed and acceleration with age across several seasons, our work offers a more detailed insight than earlier research that mostly concentrated on predictive modeling and overall performance patterns. This study places a strong emphasis on direct exploratory data analysis to find patterns that might not necessarily line up with predictive models, whereas other research, like Ma (2024) and Eid et al. (2024), mostly rely on machine learning algorithms for broad performance predictions. Furthermore, our work presents a novel approach to reducing the influence of outliers by dividing players into larger age groups—a step that hasn't been specifically discussed in earlier research we encountered. Furthermore, although Zhang et al. (2023) investigated wearable technology for continuously monitoring performance, the research incorporates past data to evaluate long-term changes in acceleration.

Ultimately, the findings highlight the complexity of the relationship between age and physical performance, suggesting that individual variability plays a significant role in defying general assumptions.

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